Islamic University of Technology

Machine Learning

CSE 4621

# Convexity

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# 1 Convexity

## 1.1 Why Should We Care?

Before we dive into the math, let us think about *why* convexity even matters in machine learning.

Imagine you are hiking in hilly terrain and you want to find the lowest point. If the terrain is shaped like a smooth bowl, no matter where you start walking downhill, you will always end up at the same bottom. That is essentially what a convex function gives us, a guarantee that there is *only one* lowest point

In machine learning, we are constantly trying to minimize some cost function $J(\theta)$. If that cost function is convex, life gets much easier as we do not have to worry about getting stuck in a local valley. Convexity is therefore one of the most desirable properties a cost function can have.

## 1.2 Convex Sets

Before talking about convex functions, we need to understand convex sets.

**Definition 1.1** (Convex Set)**.** *A set $\mathcal{C}$ is called* ***convex*** *if, for any two points $\mathbf{x}, \mathbf{y} \in \mathcal{C}$ and any $t \in [0,1]$, the point*

$$t\mathbf{x} + (1-t)\mathbf{y} \in \mathcal{C}.$$

**In plain English:** If you pick any two points inside the set and draw a straight line between them, every point on that line must also be inside the set for that set to be considered a convex set.

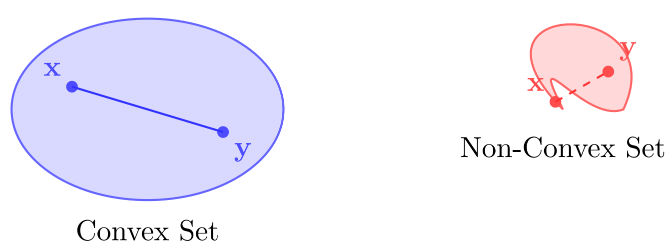


Figure 1: Left: A convex set: the line between any two points stays inside the set. Right: A non-convex set: the dashed line exits the boundary.

## 1.3 Convex Functions

A function is convex if the "line" connecting any two points on its curve sits *above* (or on) the curve itself.

**Definition 1.2** (Convex Function)**.** *A function $f : \mathbb{R}^n \to \mathbb{R}$ defined on a convex domain is* ***convex*** *if, for all* $\mathbf{x}, \mathbf{y}$ *in its domain and all* $t \in [0, 1]$*:*

$$f\big(t\mathbf{x} + (1-t)\mathbf{y}\big) \;\leq\; t f(\mathbf{x}) + (1-t) f(\mathbf{y}).$$

**Think of this graphically:** If you connect two points on the graph of $f$ with a straight line (called a *chord*), the chord must never dip below the curve.

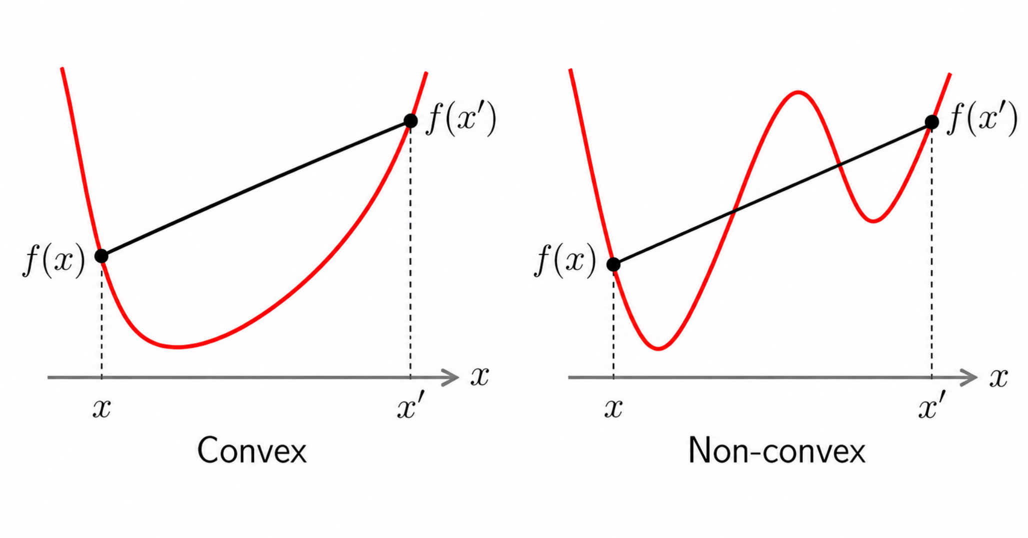


Figure 2: Left: A convex function: the line between any two points stays *above* the function. Right: A non-convex function: the line connecting any two points can go *below* the function.

**Note**

$f(x) = x^2$ is convex. Pick any two points and draw a line between them, the parabola always lies *below* that line. On the other hand, $f(x) = -x^2$ (an upside-down parabola) is not convex.

## 1.4 How to Check Convexity

Checking convexity directly from the definition can be cumbersome. Fortunately, for smooth functions (differentiable functions), there are simpler equivalent conditions.

**Theorem**

Let $f$ be a twice-differentiable function on a convex open domain $\operatorname{dom} f$. The following three statements are all **equivalent**:

**(a)** $f$ is convex.

**(b) First-order condition:** For all $\mathbf{x}, \mathbf{y}$ in the domain,

$$f(\mathbf{y}) \;\geq\; f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

This says the function always lies *above* its tangent line (or tangent hyperplane in higher dimensions).

**(c) Second-order condition:** The Hessian matrix $\nabla^2 f(\mathbf{x})$ is *positive semi-definite* (PSD) everywhere on the domain:

$$\nabla^2 f(\mathbf{x}) \succeq 0 \quad \forall \mathbf{x} \in \operatorname{dom} f.$$

In the one-dimensional case, this simply means $f''(x) \geq 0$ for all $x$ i.e., the function is "curving upwards" everywhere.

**Note**

**Intuition for the second-order condition:** The second derivative tells us whether a function is curving up or down. If $f''(x) \geq 0$ everywhere, the function is always bending *upwards* (like a bowl), which is exactly what convexity means. If $f''(x) < 0$ somewhere, the function dips downward at that point – it is not convex there.

# 2 Convexity of Mean Squared Error (MSE)

## 2.1 A Quick Recap of Linear Regression

In linear regression, we try to predict a target value $y$ from input features $\mathbf{x}$ using a linear model:

$$\hat{y} = \theta^\top \mathbf{x},$$

where $\theta$ is the vector of parameters (weights) we want to learn.

To measure how well our model fits the data, we use the **Mean Squared Error (MSE)**:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(\theta^\top \mathbf{x}^{(i)} - y^{(i)}\right)^2$$

Our goal is to find the $\theta$ that *minimizes* $J(\theta)$.

## 2.2 Is MSE Convex?

We will use the second-order condition from the previous section. We need to compute the Hessian of $J(\theta)$ with respect to $\theta$ and check whether it is positive semi-definite.

**Step 1 – First derivative (gradient):**

$$\begin{aligned}
\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) &= \nabla_{\boldsymbol{\theta}} \left[ \frac{1}{2m} \sum_{i=1}^{m} \left(\boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)}\right)^2 \right] \\
&= \frac{2}{2m} \sum_{i=1}^{m} \left(\boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)}\right) \cdot \mathbf{x}^{(i)} \\
&= \frac{1}{m} \sum_{i=1}^{m} \left(\boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)}\right) \mathbf{x}^{(i)}.
\end{aligned}$$

**Step 2 – Second derivative (Hessian):**

$$\begin{aligned}
\nabla^2_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) &= \nabla_{\boldsymbol{\theta}} \left[ \frac{1}{m} \sum_{i=1}^{m} \left(\boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)}\right) \mathbf{x}^{(i)} \right] \\
&= \frac{1}{m} \sum_{i=1}^{m} \mathbf{x}^{(i)} \left(\mathbf{x}^{(i)}\right)^\top.
\end{aligned}$$

**Step 3 – Check positive semi-definiteness:**
For any vector $\mathbf{v} \in \mathbb{R}^d$:

$$\mathbf{v}^\top \left( \frac{1}{m} \sum_{i=1}^{m} \mathbf{x}^{(i)} \left(\mathbf{x}^{(i)}\right)^\top \right) \mathbf{v} = \frac{1}{m} \sum_{i=1}^{m} \mathbf{v}^\top \mathbf{x}^{(i)} \left(\mathbf{x}^{(i)}\right)^\top \mathbf{v}$$
$$= \frac{1}{m} \sum_{i=1}^{m} \left(\mathbf{v}^\top \mathbf{x}^{(i)}\right)^2 \geq 0.$$

Since the squared norm is always non-negative, this expression is always $\geq 0$. Therefore, $\nabla^2 J(\mathbf{w}) \succeq 0$ for all $\mathbf{w}$.

**Key Takeaway**

**Conclusion:** The MSE cost function $J(\theta)$ is **convex** in the parameters $\theta$. It means that when we train a linear regression model, any minimum we find is guaranteed to be a *global* minimum.

**Note**

Notice that the Hessian does not depend on $\theta$ at all! This means $J(\theta)$ is not just a convex function, it is a **quadratic** function in $\theta$, which is one of the simplest and most well-behaved types of convex functions.

# 3 Optimality of a Convex Function

## 3.1 Local vs. Global Optima

When minimizing a function, we often talk about two kinds of "best" points:

**Definition 3.1** (Local Optimum)**.** *A point $\theta^*$ is a **local minimum** of $f$ if there exists a small neighborhood around $\theta^*$ such that*

$$f(\theta^*) \leq f(\theta) \quad \textit{for all } \theta \textit{ in that neighborhood.}$$

*In other words, it is the best point nearby, but there might be even better points far away.*

**Definition 3.2** (Global Optimum)**.** *A point $\theta^*$ is a* ***global minimum*** *of $f$ if*

$$f(\theta^*) \leq f(\theta) \quad \textit{for all } \theta \textit{ in the entire domain.}$$

*This is the best point everywhere,not just locally.*

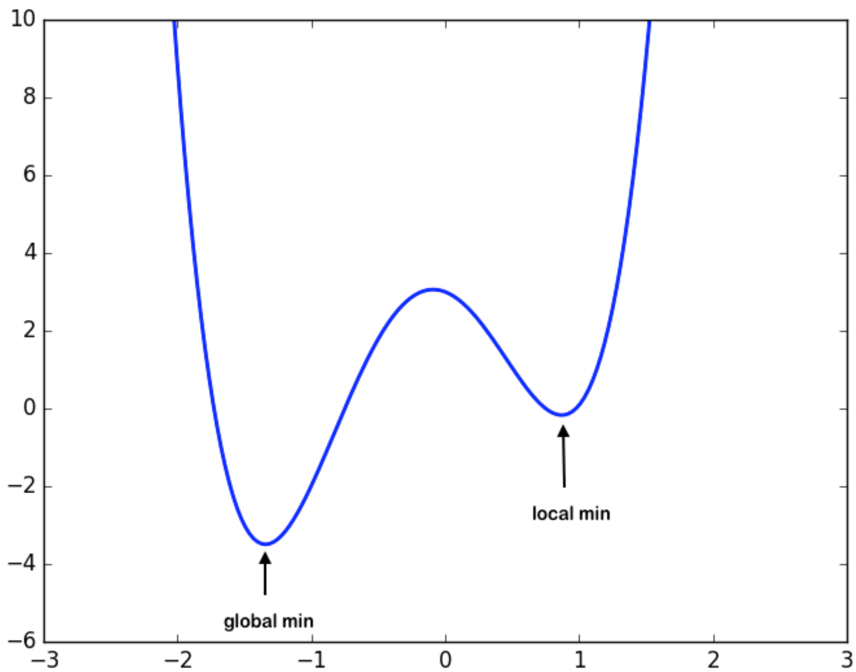


Figure 3: A non-convex function can have many local minima. Only one of them is the global minimum.

## 3.2 The Power of Convexity

For general (non-convex) functions, finding the global minimum is very hard as you might get stuck in a local minimum and never realize there is a better solution elsewhere. Convexity eliminates this problem entirely.

### Theorem

**Theorem:** If $f$ is a **convex** function, then every local minimum is also a global minimum.

**Proof:** Suppose $\boldsymbol{\theta}^*$ is a local minimum of $f$. We want to show it must also be a global minimum.

Assume, for the sake of contradiction, that $\boldsymbol{\theta}^*$ is *not* a global minimum. Then there exists some point $\mathbf{u}$ such that

$$f(\mathbf{u}) < f(\boldsymbol{\theta}^*).$$

Now consider any point on the line segment between $\boldsymbol{\theta}^*$ and $\mathbf{u}$:

$$\mathbf{z} = \boldsymbol{\theta}^* + t(\mathbf{u} - \boldsymbol{\theta}^*) = (1-t)\boldsymbol{\theta}^* + t\mathbf{u}, \quad t \in (0,1).$$

By convexity of $f$:

$$f(\mathbf{z}) \leq (1-t)f(\boldsymbol{\theta}^*) + tf(\mathbf{u}) < (1-t)f(\boldsymbol{\theta}^*) + tf(\boldsymbol{\theta}^*) = f(\boldsymbol{\theta}^*).$$

So $f(\mathbf{z}) < f(\boldsymbol{\theta}^*)$. But $\mathbf{z}$ can be made arbitrarily close to $\boldsymbol{\theta}^*$ by taking $t$ very small. This contradicts the assumption that $\boldsymbol{\theta}^*$ is a local minimum.

Therefore, no such $\mathbf{u}$ can exist, and $\boldsymbol{\theta}^*$ must be a global minimum.

### Key Takeaway

You can think of it this way: because $J(\theta)$ is shaped like a bowl, no matter where you "roll the ball" downhill, it will always end up at the same lowest point. There are no false valleys or traps, only one unique global minimum.

### Note

Many modern machine learning models; such as neural networks, have *non-convex* cost functions. In those cases, gradient descent can get stuck at local minima or saddle points. Convexity is a special, fortunate property of simpler models like linear regression, and it is one of the main reasons linear regression is both theoretically clean and practically reliable.